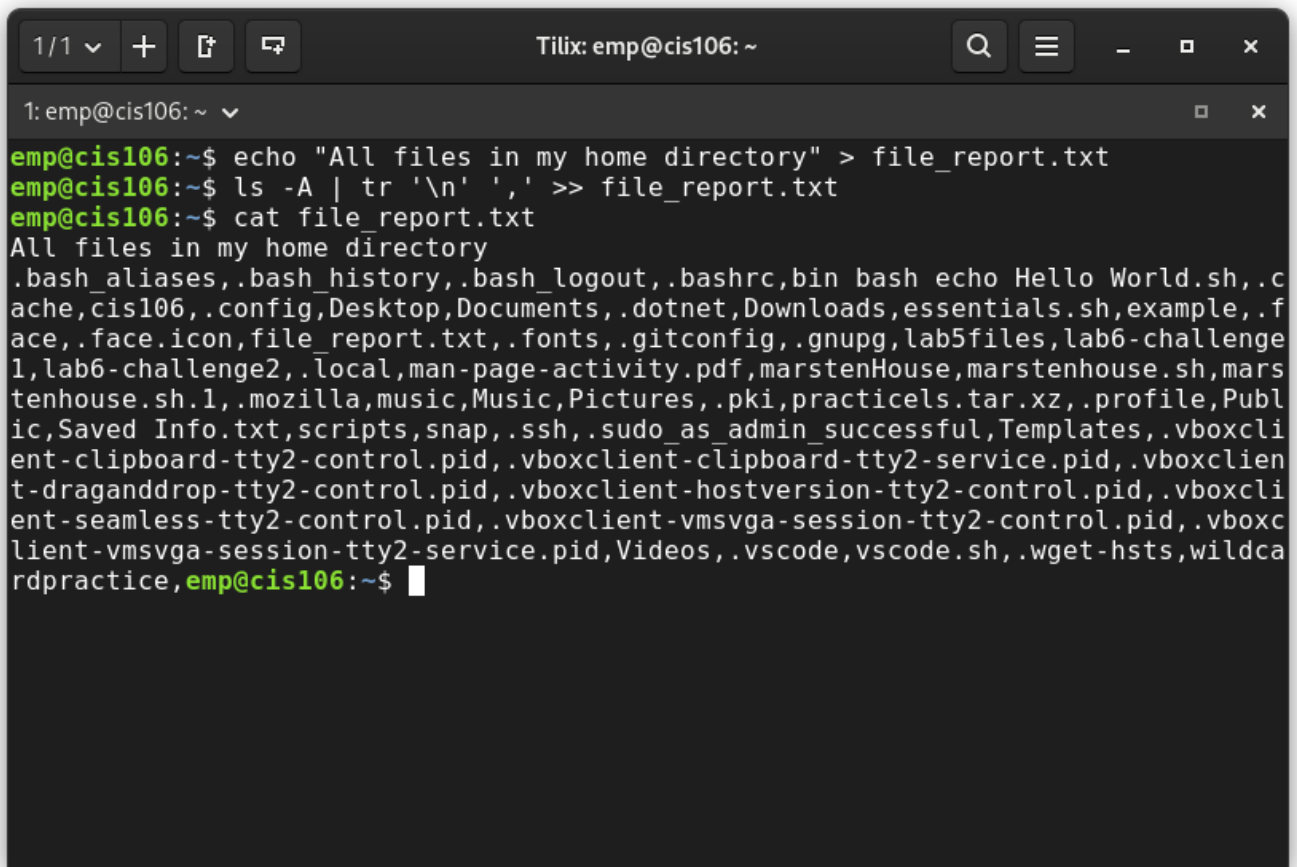


Lab 9

Question 1: Working with saving commands output to a file



```
Tilix: emp@cis106: ~  
1: emp@cis106: ~  
emp@cis106:~$ echo "All files in my home directory" > file_report.txt  
emp@cis106:~$ ls -A | tr '\n' ',' >> file_report.txt  
emp@cis106:~$ cat file_report.txt  
All files in my home directory  
.bash_aliases,.bash_history,.bash_logout,.bashrc,bin bash echo Hello World.sh,.c  
ache,cis106,.config,Desktop,Documents,.dotnet,Downloads,essentials.sh,example,.f  
ace,.face.icon,file_report.txt,.fonts,.gitconfig,.gnupg,lab5files,lab6-challenge  
1,lab6-challenge2,.local,man-page-activity.pdf,marstenHouse,marstenhouse.sh,mars  
tenhouse.sh.1,.mozilla,music,Music,Pictures,.pki,practicels.tar.xz,.profile,Publ  
ic,Saved Info.txt,scripts,snap,.ssh,.sudo_as_admin_successful,Templates,.vboxcli  
ent-clipboard-tty2-control.pid,.vboxclient-clipboard-tty2-service.pid,.vboxclien  
t-draganddrop-tty2-control.pid,.vboxclient-hostversion-tty2-control.pid,.vboxcli  
ent-seamless-tty2-control.pid,.vboxclient-vmvga-session-tty2-control.pid,.vboxc  
lient-vmvga-session-tty2-service.pid,Videos,.vscode,vscode.sh,.wget-hsts,wildca  
rdpractice,emp@cis106:~$
```

Question 2: Working with the pipe (|)

```
Tilix emp@cis106: ~  
1: emp@cis106: ~  
emp@cis106:~$ echo "Total number of files in my home directory"  
Total number of files in my home directory  
emp@cis106:~$ echo "Total number of files in my home directory" >> file_report.txt  
emp@cis106:~$ ls -la --ignore=. --ignore=.. | grep -v '^d' | grep -v '^total' | wc -l >> file_report.txt  
emp@cis106:~$ echo "Total number of directories in my home directory" >> file_report.txt  
emp@cis106:~$ ls -la --ignore=. --ignore=.. | grep '^d' | wc -l >> file_report.txt  
emp@cis106:~$ tail -n 10 file_report.txt  
All files in my home directory  
.bash_aliases,.bash_history,.bash_logout,.bashrc,bin bash echo Hello World.sh,.cache,cis106,.config,Desktop/Documents,.dotnet,Downloads,essentials.sh,example,.face,.face.icon,file_report.txt,.fonts,.gitconfig,gnuPG,lab5files,lab6-challenge1,lab6-challenge2,.local,man-page-activity.pdf,marstenHouse,marstenhouse.sh,marstenhouse.sh.1,.mozilla,music,MUSIC,Pictures,.pkc,practicels.tar.xz,.profile,Public,Saved Info.txt,scripts,snap,.ssh,.sudo_as_admin_successful,Templates,.vboxclient-clipboard-tty2-control.pid,.vboxclient-clipboard-tty2-service.pid,.vboxclient-draganddrop-tty2-control.pid,.vboxclient-hostversion-tty2-control.pid,.vboxclient-seamless-tty2-control.pid,.vboxclient-vmsvga-session-tty2-control.pid,.vboxclient-vmsvga-session-tty2-service.pid,Videos,.vscode,vscodc.sh,.wget-hsts,wxldcardpractice>Total number of files in my home directory  
26  
Total number of directories in my home directory  
28  
emp@cis106:~$
```

Question 3: awk Command

```
1/1 ~ + [ ] [ ] Tili: emp@cis106: ~  
t:emp@cis106:~  
emp@cis106:~$ awk -F: '{print $1}' /etc/passwd | head -5  
root  
daemon  
bin  
sys  
sync  
emp@cis106:~$ awk -F: '{print $1, $NF}' /etc/passwd | head -5  
root /bin/bash  
daemon /usr/sbin/nologin  
bin /usr/sbin/nologin  
sys /usr/sbin/nologin  
sync /bin/sync  
emp@cis106:~$ awk -F: '{print "The user: " $1 " logs in with: " $7 " and is home dir is: " $6}' /etc/passwd | head -8  
The user: root logs in with: /bin/bash and is home dir is: /root  
The user: daemon logs in with: /usr/sbin/nologin and is home dir is: /usr/sbin  
The user: bin logs in with: /usr/sbin/nologin and is home dir is: /bin  
The user: sys logs in with: /usr/sbin/nologin and is home dir is: /dev  
The user: sync logs in with: /bin/sync and is home dir is: /bin  
The user: games logs in with: /usr/sbin/nologin and is home dir is: /usr/games  
The user: man logs in with: /usr/sbin/nologin and is home dir is: /var/cache/man  
The user: lp logs in with: /usr/sbin/nologin and is home dir is: /var/spool/lpd  
emp@cis106:~$ awk -F: '{print "The user: " $1 " logs in with: " $7 " and is home dir is: " $6}' /etc/passwd > report_users.txt  
emp@cis106:~$ awk -F: '{ $7 == "/bin/bash" {print $1 " = " $7} }' /etc/passwd >> report_users.txt  
emp@cis106:~$ cat report_users.txt  
The user: root logs in with: /bin/bash and is home dir is: /root  
The user: daemon logs in with: /usr/sbin/nologin and is home dir is: /usr/sbin  
The user: bin logs in with: /usr/sbin/nologin and is home dir is: /bin  
The user: sys logs in with: /usr/sbin/nologin and is home dir is: /dev  
The user: sync logs in with: /bin/sync and is home dir is: /bin  
The user: games logs in with: /usr/sbin/nologin and is home dir is: /usr/games  
The user: man logs in with: /usr/sbin/nologin and is home dir is: /var/cache/man  
The user: lp logs in with: /usr/sbin/nologin and is home dir is: /var/spool/lpd  
The user: mail logs in with: /usr/sbin/nologin and is home dir is: /var/mail  
The user: news logs in with: /usr/sbin/nologin and is home dir is: /var/spool/news  
The user: uucp logs in with: /usr/sbin/nologin and is home dir is: /var/spool/uucp  
The user: proxy logs in with: /usr/sbin/nologin and is home dir is: /bin  
The user: www-data logs in with: /usr/sbin/nologin and is home dir is: /var/www  
The user: backup logs in with: /usr/sbin/nologin and is home dir is: /var/backups  
The user: list logs in with: /usr/sbin/nologin and is home dir is: /var/list  
The user: irc logs in with: /usr/sbin/nologin and is home dir is: /run/ircd
```

Question 4: Working with CSV files

```
Tilix: emp@cis106: ~  
1/1 ▾ + ⌵ 🔍  
1: emp@cis106: ~  
emp@cis106:~$ head -n 1 cars.csv > hondal.csv  
emp@cis106:~$ grep -i "honda" cars.csv >> hondal.csv  
emp@cis106:~$ cat hondal.csv  
Make,Model,Type,Origin,DriveTrain,MSRP,Invoice,EngineSize,Cylinders,Horsepower  
Honda,Civic, Sedan,Asia,Front,18990,17300,1.8,4,140  
Honda,Accord, Sedan,Asia,Front,22990,21000,2.4,4,185  
Honda,CR-V, SUV,Asia,All,25990,23800,2.4,4,160  
Honda,Pilot, SUV,Asia,All,31990,29200,3.5,6,240  
Honda,Odyssey, Minivan,Asia,Front,28990,26500,3.5,6,255  
emp@cis106:~$ sed 's/,;/g' hondal.csv > honda2.csv  
emp@cis106:~$ cat honda2.csv  
Make;Model;Type;Origin;DriveTrain;MSRP;Invoice;EngineSize;Cylinders;Horsepower  
Honda;Civic; Sedan;Asia;Front;18990;17300;1.8;4;140  
Honda;Accord; Sedan;Asia;Front;22990;21000;2.4;4;185  
Honda;CR-V; SUV;Asia;All;25990;23800;2.4;4;160  
Honda;Pilot; SUV;Asia;All;31990;29200;3.5;6;240  
Honda;Odyssey; Minivan;Asia;Front;28990;26500;3.5;6;255  
emp@cis106:~$ echo "Car,Cylinders,Horsepower" > honda3.csv  
emp@cis106:~$ awk -F, 'NR>1 {print $1 "," $9 "," $10}' honda2.csv >> honda3.csv  
emp@cis106:~$ sed 's"/"/g' honda3.csv > honda3_final.csv  
emp@cis106:~$ mv honda3_final.csv honda3.csv  
emp@cis106:~$ cat honda3.csv  
Car,Cylinders,Horsepower  
Honda,Civic, Sedan,Asia,Front,18990,17300,1.8,4,140,,  
Honda,Accord, Sedan,Asia,Front,22990,21000,2.4,4,185,,  
Honda,CR-V, SUV,Asia,All,25990,23800,2.4,4,160,,,  
Honda,Pilot, SUV,Asia,All,31990,29200,3.5,6,240,,,  
Honda,Odyssey, Minivan,Asia,Front,28990,26500,3.5,6,255,,  
emp@cis106:~$
```

Challenge Question: Fixing the Broken Website

Directory Structure:

```

1/1 + TiliX: emp@cis106: ~/broken
emp@cis106:~/broken$ wget https://cis106.com/assets/lab8/broken.zip
--2026-04-28 23:49:16-- https://cis106.com/assets/lab8/broken.zip
Resolving cis106.com (cis106.com)... 185.199.108.153
Connecting to cis106.com (cis106.com)[185.199.108.153]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1078606 (1.0M) [application/x-zip-compressed]
Saving to: 'broken.zip'

broken.zip                               100%[=====] 1.03M  --.-KB/s  in 0.09s

2026-04-28 23:49:16 (11.0 MB/s) - 'broken.zip' saved [1078606/1078606]

emp@cis106:~/broken$ unzip broken.zip
Archive: broken.zip
  creating: broken/
  inflating: broken/favicon.ico
  inflating: broken/profile.png
  inflating: broken/contact.html
  inflating: broken/styles.css
  inflating: broken/index.html
  inflating: broken/scripts.js
  inflating: broken/projects.html
  inflating: broken/resume.html
emp@cis106:~/broken$ cd broken
emp@cis106:~/broken$ ls -la
total 1360
drwx----- 2 emp emp 4096 Apr  9 2024 .
drwx----- 31 emp emp 4096 Apr 28 23:49 ..
-rw-rw-r-- 1 emp emp 9275 Mar 25 2023 contact.html
-rw-rw-r-- 1 emp emp 23462 Mar 25 2023 favicon.ico
-rw-rw-r-- 1 emp emp 27570 Mar 25 2023 index.html
-rw-rw-r-- 1 emp emp 1033963 Mar 25 2023 profile.png
-rw-rw-r-- 1 emp emp 6395 Mar 25 2023 projects.html
-rw-rw-r-- 1 emp emp 15143 Mar 25 2023 resume.html
-rw-rw-r-- 1 emp emp 325 Mar 25 2023 scripts.js
-rw-rw-r-- 1 emp emp 252666 Mar 25 2023 styles.css
emp@cis106:~/broken$ mkdir -p styles scripts
emp@cis106:~/broken$ mv styles.css styles/ 2>/dev/null
emp@cis106:~/broken$ mv script.js scripts/ 2>/dev/null
emp@cis106:~/broken$ ls -R
.:
contact.html  favicon.ico  index.html  profile.png  projects.html  resume.html  scripts  scripts.js  styles

./scripts:

./styles:
styles.css
emp@cis106:~/broken$ grep -E 'href|src' index.html
<link rel="icon" type="image/x-icon" href="assets/favicon.ico" />
<link rel="preconnect" href="https://fonts.googleapis.com" />
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />
<link href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@100;200;300;400;500;600;700;800;900&display=swap" rel="stylesheet" />
<link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.8.1/font/bootstrap-icons.css" rel="stylesheet" />
<link href="css/styles.css" rel="stylesheet" />
    <a class="navbar-brand" href="index.html"><span class="fw-bolder text-primary">Start Bootstrap</span></a>
    <li class="nav-item"><a class="nav-link" href="index.html">Home</a></li>
    <li class="nav-item"><a class="nav-link" href="resume.html">Resume</a></li>
    <li class="nav-item"><a class="nav-link" href="projects.html">Projects</a></li>
    <li class="nav-item"><a class="nav-link" href="contact.html">Contact</a></li>
    <a class="btn btn-primary btn-lg px-5 py-3 me-sm-3 fs-6 fw-bolder" href="resume.html">Resume</a>
    <a class="btn btn-outline-dark btn-lg px-5 py-3 fs-6 fw-bolder" href="projects.html">Projects</a>
    
    <a class="text-gradient" href="#"><i class="bi bi-twitter"></i></a>
    <a class="text-gradient" href="#"><i class="bi bi-linkedin"></i></a>
    <a class="text-gradient" href="#"><i class="bi bi-github"></i></a>
    <a class="small" href="#">Privacy</a>
    <a class="small" href="#">Terms</a>
    <a class="small" href="#">Contact</a>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.2.3/dist/js/bootstrap.bundle.min.js"></script>
<script src="js/scripts.js"></script>
emp@cis106:~/broken$ echo "Opening in Firefox..."
Opening in Firefox...
emp@cis106:~/broken$ firefox index.html &
[1] 5363
emp@cis106:~/broken$

```

Working Website in Firefox:

